一、

(a) $\because \int_{-1}^{1} f(x)\,dx = \int_{-1}^{1} \frac{c}{\sqrt{1-x^2}} = 1$

即 $c\pi = 1$, $\therefore c = \frac{1}{\pi}$

(b) $\therefore f(-\frac{1}{2} < X < \frac{1}{2}) = \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{c}{\sqrt{1-x^2}}\,dx$

$= \frac{1}{3}$

即 $P = \frac{1}{3}$

五. $\because f(-x) = f(x)$

(a) $\therefore F(-a) = \int_{-\infty}^{-a} f(x)\,dx$

$= 1 - \int_{-a}^{+\infty} f(x)\,dx$

$= 1 - \int_{-a}^{+\infty} f(-x)\,dx$

$= 1 - \int_{-\infty}^{+a} f(x)\,dx = 1 - F(a)$

且 $F(-a) = \frac{1}{2} - \int_{-a}^{0} f(x)\,dx$

$= \frac{1}{2} - \int_{0}^{a} f(-x)\,d(-x) = \frac{1}{2} - \int_{0}^{a} f(x)\,dx)$

(b) $P(|X| < a) = F(a) - F(-a)$

$= 2F(a) - 1.$

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(c) $P(|X|>a) = 1 - P(|X|<a)$

$= 2 - 2F(a)$

七. $\because f(X>10) = \frac{30-10}{30} = \frac{2}{3}$

而 $f(X>25 \mid X>15) = \frac{30-25}{30-15} = \frac{1}{3}$

九. $\because f(x) = \frac{1}{241} e^{-\frac{1}{241}x}$;

$\therefore F(T) = \begin{cases} 0 & T<0 \\ 1-e^{-\frac{1}{241}T} & T\geq 0. \end{cases}$

$\therefore P(50<T<100) = F(100) - F(50) = 0.152$

十. $\because \lambda = \frac{1}{5}$

$\therefore F(T) = \begin{cases} 0 & T<0 \\ 1-e^{-\frac{1}{5}T} & \end{cases}$

$\therefore P(\text{离开}) = e^{-2}$;

连续等红灯5次

Y	0	1	2	3	4	5
P(Y)	$(1-e^{-2})^5$	$(1-e^{-2})^4 \cdot e^{-2} \cdot 5$	$(1-e^{-2})^3 \cdot e^{-4} \cdot 10$	$(1-e^{-2})^2 \cdot e^{-6} \cdot 10$	$5(1-e^{-2}) \cdot e^{-8}$	e^{-10}

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$\therefore E(Y) = 5 \cdot e^{-2}$

$P(Y \geq 1) = 1 - (1 - e^{-2})^5$

十一.

$\because X \sim N(2, \sigma^2)$，且 $F(4) - F(2) = 0.3$

$\therefore F(4) - F(0) = 0.6$

$\therefore F(0) = \frac{1}{2} - (F(2) - F(0))$

$= 0.5 - 0.3 = 0.2$

十二.

$\therefore E(X) = \int_{-\infty}^{+\infty} x f(x)\, dx$

$= e^2 \sqrt{\pi} \cdot \frac{1}{\sqrt{\pi}} = e^2$

$E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x)\, dx$

$= \frac{3}{2} e^2 \sqrt{\pi} \cdot \frac{1}{\sqrt{\pi}} = \frac{3}{2} e^2$

$\therefore D(X) = E(X^2) - E^2(X)$

$= \frac{3}{2} e^4 - e^4$

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十四.

$\because X \sim N(\mu, \sigma^2)$

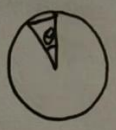
$\therefore E(X-\mu)=0;$

$$\therefore E(|X-\mu|) = 2\cdot\int_{-\infty}^{u}(u-x)f(x)\,dx$$

$$= 2\cdot\int_{u}^{\infty}(x-u)\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{(x-u)^2}{2\sigma^2}}\,dx$$

$$= \frac{2}{\sqrt{2\pi}}$$

十六.



$\therefore l = 2\cdot R\cdot\theta\sin\theta$

且 $\int_0^{\pi} d\theta = \pi$

$$\therefore l = \int_0^{\pi} 2R\cdot\sin\theta\cdot d\theta\cdot\frac{1}{\pi} = \frac{4R}{\pi}$$

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